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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE304 - ANALOG SIGNAL PROCESSING

20EE304 - ANALOG ELECTRONIC CIRCUITS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Design a capacitor coupled voltage follower circuit operating with $\pm 15V$ supply. Assume lower cutoff frequency $f_L = 100Hz$ and load resistance is $10k\Omega$. Use 741opamp.	8	L*3	1	1,2,3,5,9
b)	Show the operation of MOSFET as a switch	6	L2	2	1,2,3,5,9
c)	Outline the construction and operation of precision full wave rectifier using Op-amps.	6	L2	1	1,2,3,5,9
2. a)	Draw the diagram for an Op-amp capacitor-coupled Voltage Follower. Explain the circuit operation and calculate the voltage gain for given resistor values.	6	L2	1	1,2,3,5,9
b)	Sketch an Op-amp integrating circuit. Draw typical input & output wave-forms, explain the circuit operation.	8	L2	2	1,2,3,5,9
c)	Consider an NMOS fabricated with $0.18Nm$ process parameters of the MOSFET are; $L = 0.18Nm$, $w = 2Nm$, $C_{ox} = 8.6f F/Nm^2$, $N_n = 400cm^2/V_s$ and $V_{th} = 0.5v$. Find V_{Gs} and V_{Ds} that result in MOSFET operating in saturation with $I_D = 100NA$.	6	L2	2	1,2,3,5,9
3. a)	Draw the diagram for an Op-amp inverting amplifier circuit. Explain the circuit operation and calculate the voltage gain for given resistor values.	8	L2	1	1,2,3,5,9
b)	Sketch an Op-amp non-inverting amplifier circuit. Explain the circuit operation and calculate the voltage gain.	6	L2	1	1,2,3,5,9
c)	Show the operation of MOSFET as a linear amplifier.	6	L2	2	1,2,3,5,9
Unit – II					
4. a)	Explain the transconductance of a MOSFET & derive an expression for the same.	6	L3	3	1,2,3,5,9
b)	A CS amplifier has following circuit parameters. $I_D = 0.1mA$, $g_m = 0.1mS$, $R_D = 10k\Omega$, $r_o = 200k\Omega$, $R_{sig} = 100 k\Omega$, $R_G = 1M\Omega$ and load resistance $R_L = 10k\Omega$. Find amplifier parameters R_{in} , A_{v_o} , A_v , R_o and G_v .	8	L3	4	1,2,5,9
c)	Elaborate DC biasing of MOSFET using a constant current source.	6	L1	3	1,2,3,5,9
5. a)	With the neat sketch explain the working of common source amplifier connected with a source resistor. Also show its T-equivalent circuit model.	6	L2	4	1,2,5,9
b)	Analyze biasing in MOS amplifier circuit using Fixed VGS method.	6	L3	3	1,2,3,5,9
c)	Define MOSFET's unity gain frequency & determine the expression for the same.	8	L3	4	1,2,5,9

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| 6. a) | Discuss biasing scheme for a n-channel MOSFET by fixing V_G and in connection with a resistor in the source terminal. | 8 | L2 | 3 | 1,2,3,5,9 |
| b) | Describe the working of MOSFET as a small signal amplifier. | 6 | L2 | 4 | 1,2,5,9 |
| c) | Obtain small signal T-equivalent circuit of MOSFET. | 6 | L4 | 3 | 1,2,3,5,9 |

Unit – III

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| 7. a) | Draw the block diagram of a general feedback structure and hence derive an expression for the closed loop gain. | 8 | L2 | 5 | 1,2,3,5,9 |
| b) | Explain small signal analysis of MOS differential pair and derive the expression for differential gain. | 12 | L3 | 5 | 1,2,3,5,9 |
| 8. a) | Discuss the four basic topologies of feedback amplifiers. | 10 | L2 | 5 | 1,2,3,5,9 |
| b) | With a neat circuit diagram and relevant equations explain FET based RC phase shift oscillator. Mention its advantages and disadvantages. | 10 | L2 | 5 | 1,2,3,5,9 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
